

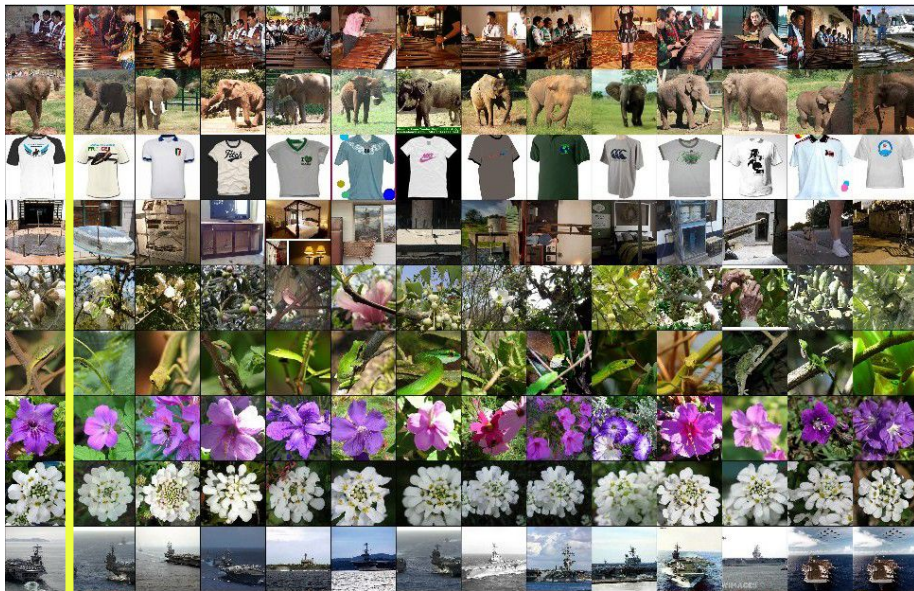
Representation learning

Deep Learning @ HSE

Plan

- Examples
- Differences to other tasks
- Classification
- Metric learning
- Self-supervised learning
- CLIP

Retrieval using learned representations



[Krizhevsky et al. NIPS12]

Verification problems in vision

Key question: do two photos show the same object/subject? (*verification*)

Face recognition
datasets (e.g.
MSRA-CF):



Re-identification
datasets (e.g.
ViPER):



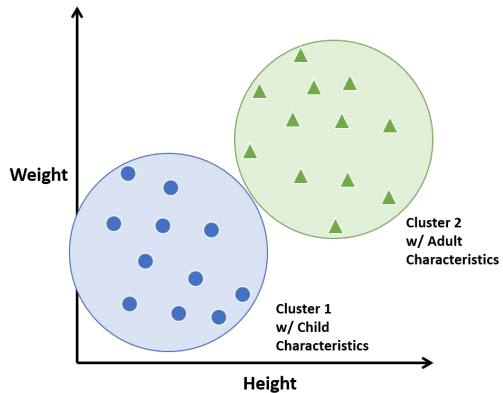
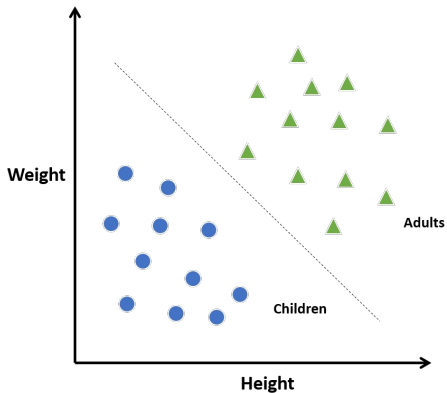
Class is important



Examples

- Retrieve similar to request
- few-shot classification
 - Rare classes
 - Enhance time-to-market
 - Non-DS can use as wrapped instrument
- Build strong embedder
- Identification
 - Faces
 - Logos
 - Calligraphy
- Re-Identification

Classification vs Clustering



Classification vs Clustering



Verification vs Classification

Key question: do two photos show the same object/subject? (*verification*)

- System must be able to handle unseen “classes”
- During training classes can be numerous, small-sized, imbalanced, etc.



Datasets

Many classes, few objects pre class

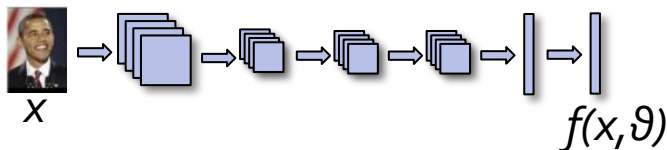
- Omniglot
- Person ReID
- LFW



Datasets. ReID

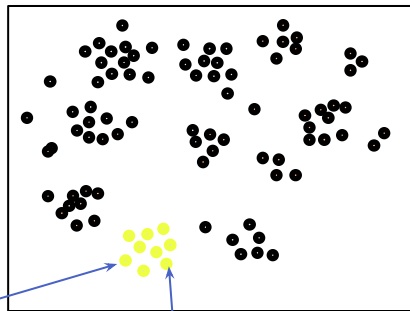


Verification as embedding learning

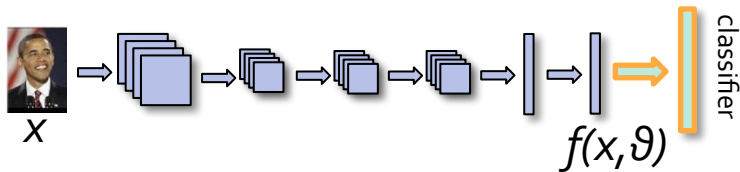


Semantic space:

NB: always
normalize your
descriptors!



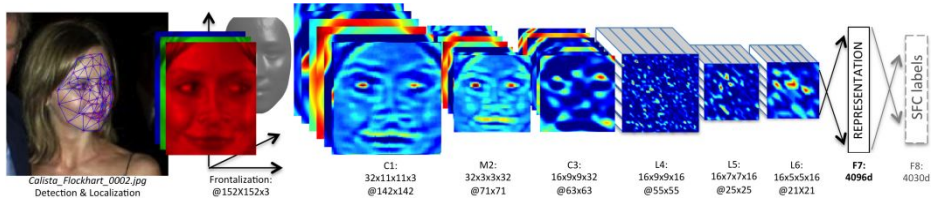
Approach 1: classification-based



- Same idea as “Train on ImageNet, use for retrieval”
- The bigger the classification dataset, the better is the performance
- Training-time classes can be seen as prototypes for test-time classes

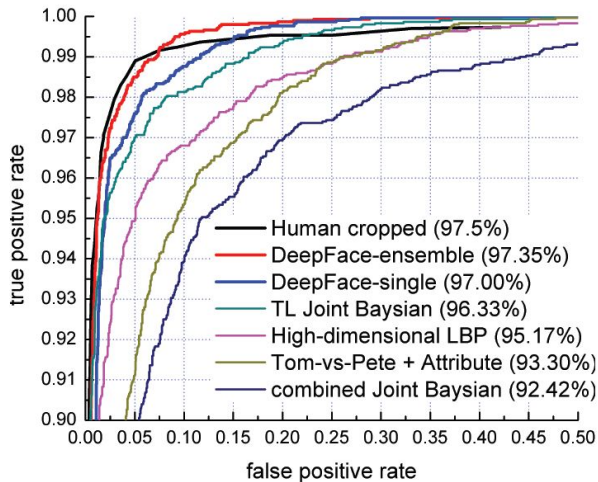
Face verification: “Deep face”

[Taigman et al. 2014]



- *Classification* network trained on 4030 people x ~1000 images.
- Target problem: *verification* (same vs different)

Face verification: “Deep face”

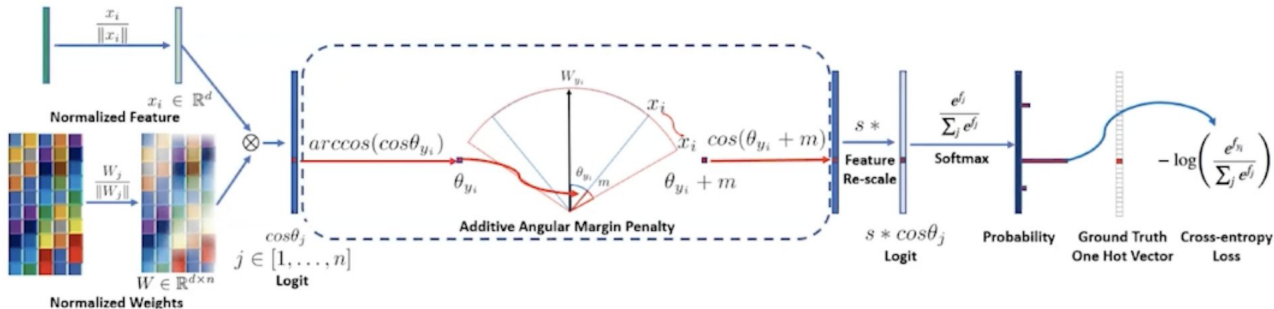


Different ConvNets combined using SVM-learned weights on validation set

[Taigman et al. 2014]

ArcFace

Angular soft-max with margin loss:



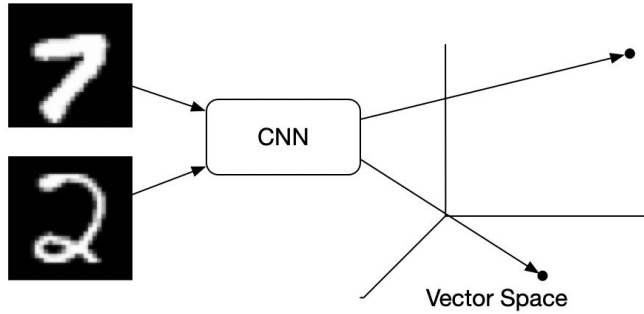
$$L_3 = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j=1, j \neq y_i}^n e^{s \cos \theta_j}}$$

[Deng et al. CVPR 2019]

From classification to metric learning

- Classification requires class labels
- Metric learning requires only same-different labels
- Classification dataset can be trivially converted to metric learning dataset

Contrastive Loss



Contrastive Loss

$$y_{\text{pred}} = \text{dist}(x_i, x_j)$$
$$y_{\text{target}} = \begin{cases} 1, & \text{if } c_i == c_j, \\ 0, & \text{otherwise} \end{cases}$$

Contrastive Loss

$$y_{\text{pred}} = \text{dist}(x_i, x_j)$$

$$y_{\text{target}} = \begin{cases} 1, & \text{if } c_i == c_j, \\ 0, & \text{otherwise} \end{cases}$$

$$L = y_{\text{target}} y_{\text{pred}} - (1 - y_{\text{target}}) y_{\text{pred}}$$

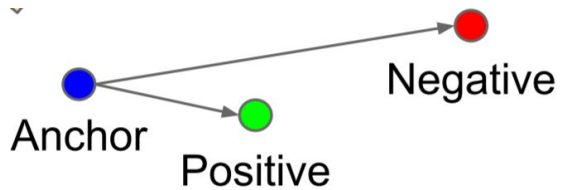
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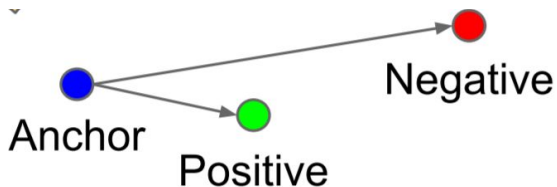
$$y_{\text{target}} = \begin{cases} 1, & \text{if } c_i == c_j, \\ 0, & \text{otherwise} \end{cases}$$

$$L = y_{\text{target}}y_{\text{pred}} + (1 - y_{\text{target}}) \max(0, m - y_{\text{pred}})$$

Triplet Loss

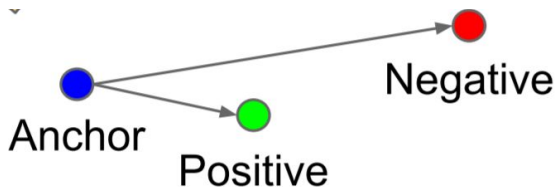


Triplet Loss



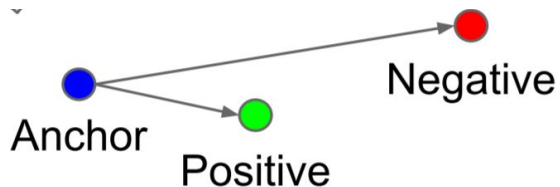
$$L = \text{dist}(x_a, x_p) - \text{dist}(x_a, x_n)$$

Triplet Loss



$$L = \text{dist}(x_a, x_p) + \max(0, m - \text{dist}(x_a, x_n))$$

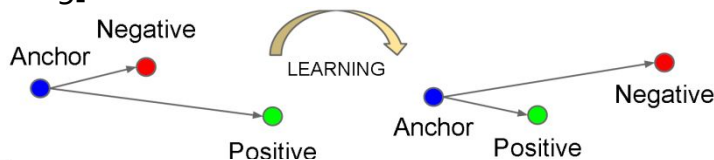
Triplet Loss + Classification



$$L = L_{\text{triplet}} + L_{\text{CE}}$$

Google “FaceNet”

[Schroff et al. CVPR15]



Simple **triplet loss**:
$$\sum_i^N \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha \right]_+$$

- Use large mini-batches (1800, 40 images for several classes + lots of random)
- Take all positives from the batch
- Mine “*semi-hard*” negatives

$$\|f(x_i^a) - f(x_i^p)\|_2^2 < \|f(x_i^a) - f(x_i^n)\|_2^2$$

Google "FaceNet" results



[Schroff et al. CVPR15]

- Upto 99.63% on LFW (human is ~97%)

Performance vs training data:

#images	VAL
2.6M	76.3%
26M	85.1%
52M	85.1%
260M	86.2%

Sampling

- All Triplet
- Hard Triplet
- Hard Cluster Triplet
- And others

All Sampling

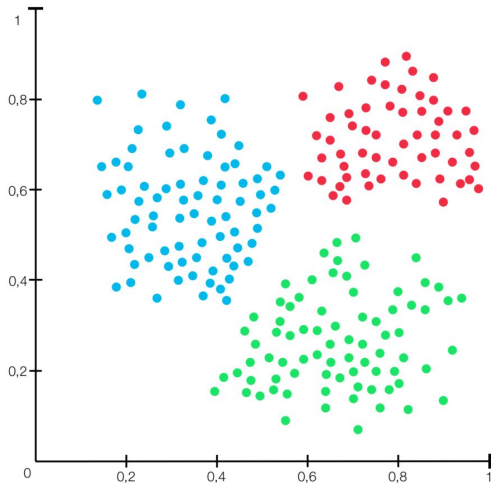
$$\mathcal{L}_{\text{BA}}(\theta; X) = \overbrace{\sum_{i=1}^P \sum_{a=1}^K}^{\text{all anchors}} \overbrace{\sum_{\substack{p=1 \\ p \neq a}}^K}^{\text{all pos.}} \overbrace{\sum_{\substack{j=1 \\ j \neq i}}^P \sum_{n=1}^K}^{\text{all negatives}} \left[m + d_{j,a,n}^{i,a,p} \right]_+, \quad (6)$$

$$d_{j,a,n}^{i,a,p} = D(f_\theta(x_a^i), f_\theta(x_p^i)) - D(f_\theta(x_a^i), f_\theta(x_n^j)).$$

Hard Sampling

$$\mathcal{L}_{\text{BH}}(\theta; X) = \sum_{i=1}^{\overbrace{P}^{\text{all anchors}}} \sum_{a=1}^{\overbrace{K}^{\text{hardest positive}}} \left[m + \max_{p=1 \dots K} D(f_{\theta}(x_a^i), f_{\theta}(x_p^i)) \right. \quad (5)$$
$$\left. - \underbrace{\min_{\substack{j=1 \dots P \\ n=1 \dots K \\ j \neq i}} D(f_{\theta}(x_a^i), f_{\theta}(x_n^j))}_{\text{hardest negative}} \right]_+,$$

Hard Cluster Sampling



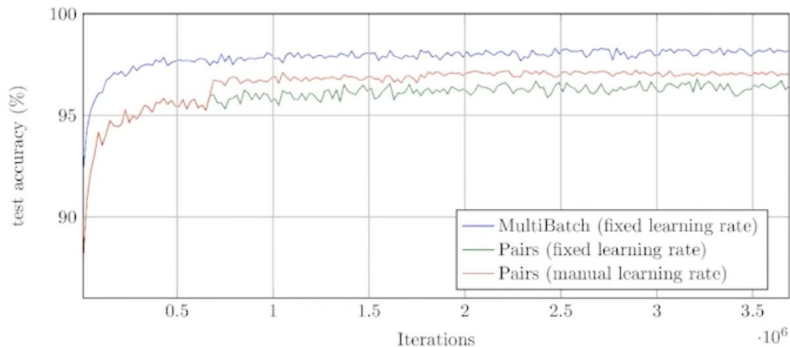
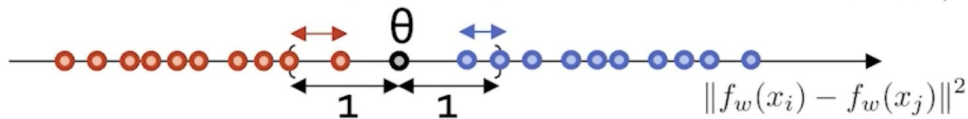
$$d_i^{inter} = \min_{\forall i_d \in P, i_d \neq i} \|f_i^m - f_{i_d}^m\|_2^2$$

$$d_i^{intra} = \max_K \|f(x) - f_i^m\|_2^2$$

$$Lb_c = \sum_i^P \max((d_i^{intra} - d_i^{inter} + \alpha), 0)$$

Quadruplet loss

$$l(w, \theta; x_i, x_j, y_{ij}) = (1 - y_{ij} (\theta - \|f_w(x_i) - f_w(x_j)\|^2))_+$$



[Tadmor et al. NIPS16]

Self-supervised representation learning

- (Pre) trains representations without labels
- Especially useful for new domains with few labels (e.g. medical, robotics)
- Meta-idea 1: bottleneck learning (reduction to compression)
- Meta-idea 2: predictive learning (reduction to classification)
- Meta-idea 3: reduction to metric learning

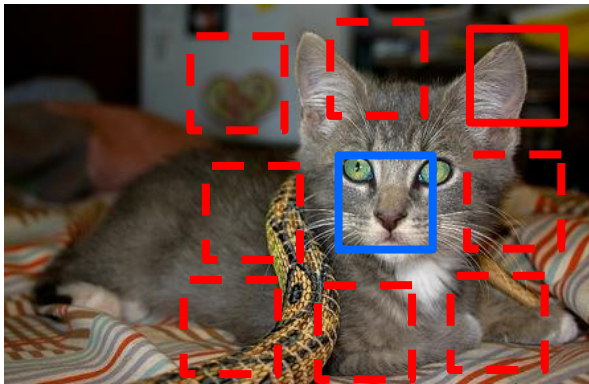
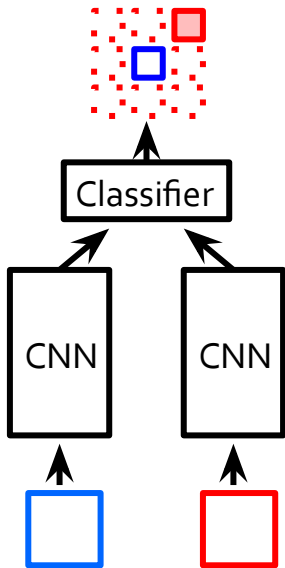
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Predictive self-supervised learning

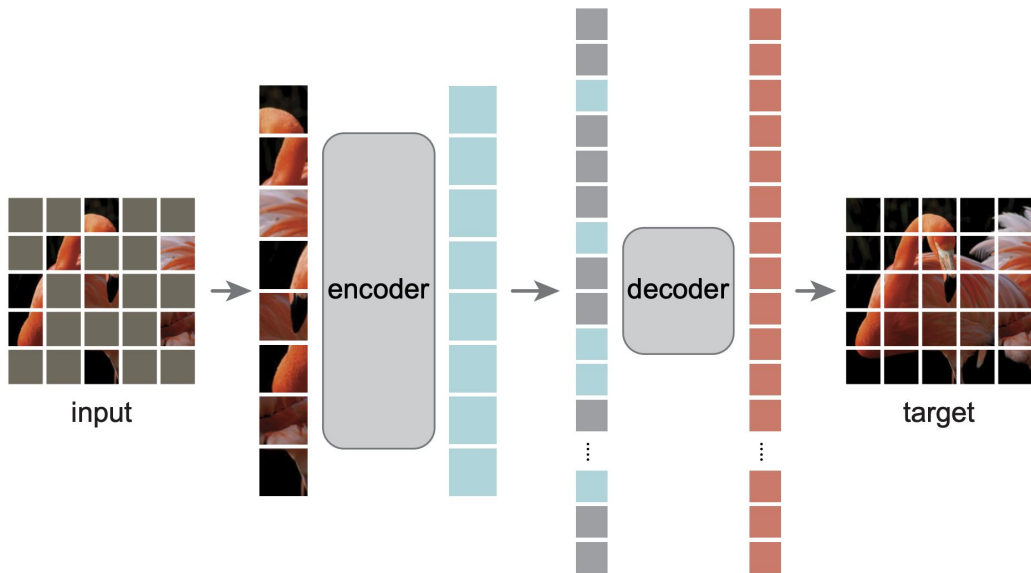
- Given an element predict *nearby* elements (e.g. next, previous, adjacent, etc.)
- Does not require annotated data (“self-supervised”)
- Particularly prominent in NLP (e.g. BERT), but also applicable to computer vision

Predictive learning for still images

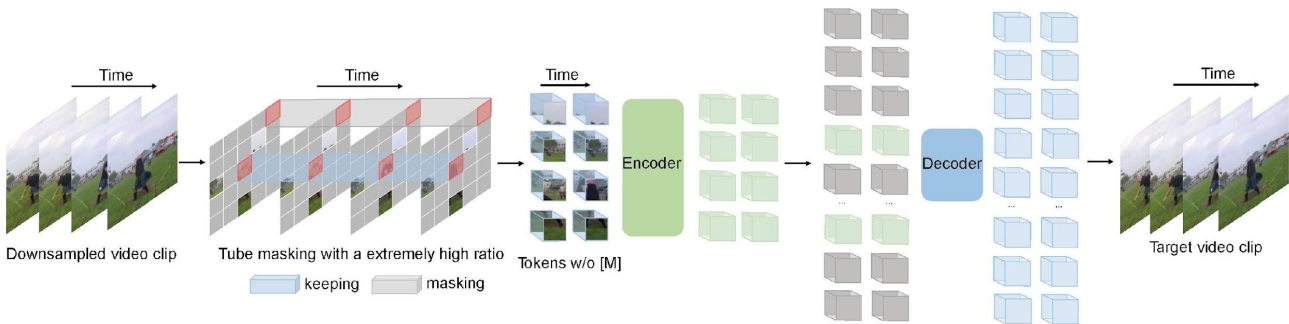


[Doersch et al. ICCV15]

Predictive learning for still images



Predictive learning for still images



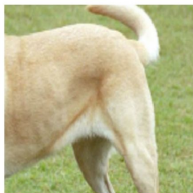
Self-supervised representation learning

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SimCLR self-supervised learning



(a) Original



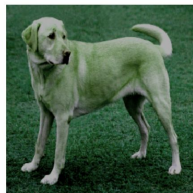
(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)

- Each batch consists of multiple pairs of matching images
- Each pair is the two copies of the same image with varying crop, color distortion, Gaussian blur
- We want the network to map each pair to two nearby points that are separated from remaining images

- Loss:

$$\ell_{i,j}^{\text{NT-Xent}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$$

- Requires large enough batch to work well

[Chen et al. ICML20] [Chen et al. NeurIPS20]

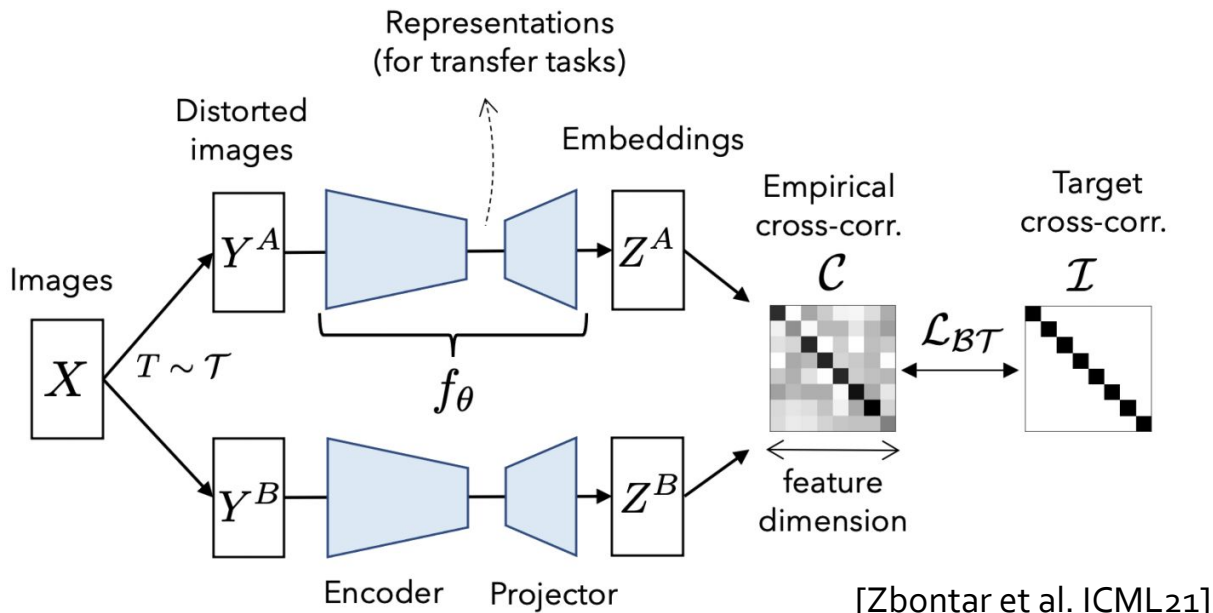
SimCLR self-supervised learning

<https://ai.googleblog.com/2020/04/advancing-self-supervised-and-semi.html>

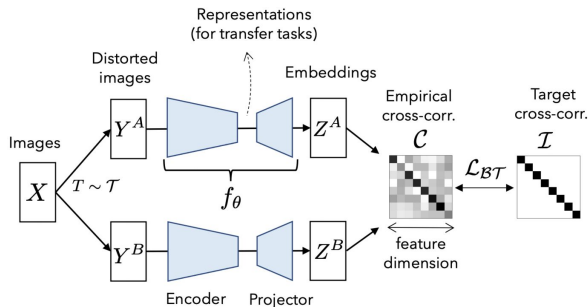
[Chen et al. ICML20]

[Chen et al. NeurIPS20]

Barlow Twins



Barlow Twins

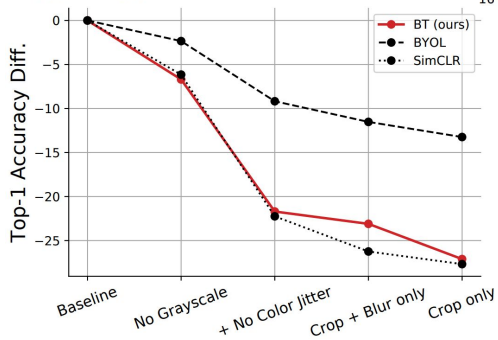
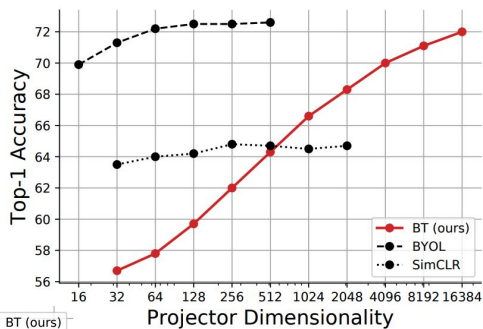
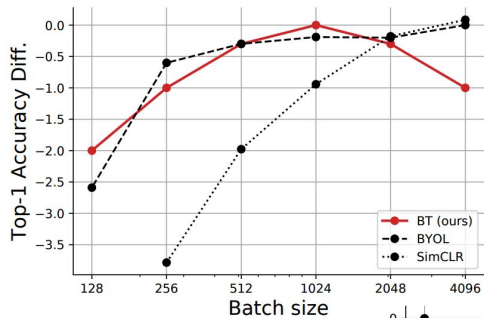


$$\mathcal{L}_{BT} \triangleq \underbrace{\sum_i (1 - C_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} C_{ij}^2}_{\text{redundancy reduction term}}$$

$$C_{ij} \triangleq \frac{\sum_b z_{b,i}^A z_{b,j}^B}{\sqrt{\sum_b (z_{b,i}^A)^2} \sqrt{\sum_b (z_{b,j}^B)^2}}$$

[Zbontar et al. ICML21]

Barlow Twins



[Zbontar et al. ICML21]

Barlow Twins

Linear classification:

Method	Places-205	VOC07	iNat18
Supervised	53.2	87.5	46.7
SimCLR	52.5	85.5	37.2
MoCo-v2	51.8	<u>86.4</u>	38.6
SwAV (w/o multi-crop)	52.8	<u>86.4</u>	39.5
SwAV	<u>56.7</u>	<u>88.9</u>	<u>48.6</u>
BYOL	<u>54.0</u>	<u>86.6</u>	<u>47.6</u>
BARLOW TWINS (ours)	<u>54.1</u>	86.2	<u>46.5</u>

Detection / instance segmentation:

Method	VOC07+12 det			COCO det			COCO instance seg		
	AP _{all}	AP ₅₀	AP ₇₅	AP ^{bb}	AP ^{bb} ₅₀	AP ^{bb} ₇₅	AP ^{mk}	AP ^{mk} ₅₀	AP ^{mk} ₇₅
Sup.	53.5	81.3	58.8	38.2	58.2	41.2	33.3	54.7	35.2
MoCo-v2	57.4	82.5	64.0	39.3	58.9	42.5	34.4	55.8	36.5
SwAV	56.1	82.6	62.7	38.4	58.6	41.3	33.8	55.2	35.9
SimSiam	57	82.4	63.7	39.2	59.3	42.1	34.4	56.0	36.7
BT (ours)	56.8	82.6	63.4	39.2	59.0	42.5	34.3	56.0	36.5

[Zbontar et al. ICML21]

Medical imaging case

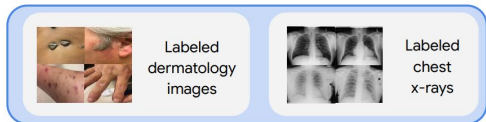
(1) Self-supervised learning on **unlabeled** natural images



(2) Self-supervised learning on **unlabeled** medical images and **Multi-Instance Contrastive Learning (MICLe)** if multiple images of each medical condition are available

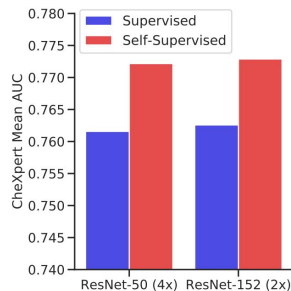
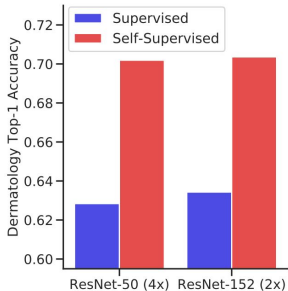


(3) Supervised fine-tuning on **labeled** medical images



15 000 train cases
(1-6 images each),
454 000 unlabeled,
27 classes

67 000 train images,
112 000 unlabeled,
5 labels



Blue = pretrain on ImNet

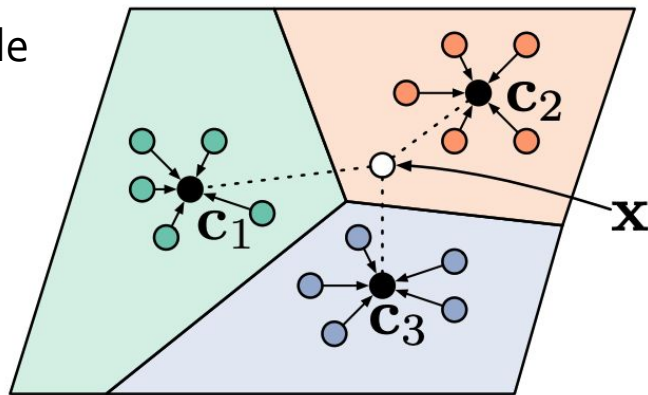
Red = unsupervised pretrain on
ImNet+Target domain

[Azizi et al. ArXiv 21]

Prototypical networks

Class is represented by a single point in the mean space:

$$\mathbf{c}_k = \frac{1}{|S_k|} \sum_{(\mathbf{x}_i, y_i) \in S_k} f_\phi(\mathbf{x}_i)$$



Classifying new examples:

$$p_\phi(y = k | \mathbf{x}) = \frac{\exp(-d(f_\phi(\mathbf{x}), \mathbf{c}_k))}{\sum_{k'} \exp(-d(f_\phi(\mathbf{x}), \mathbf{c}_{k'}))}$$

[Snell et al. ICLR17]

Meta-learning

```
 $V \leftarrow \text{RANDOMSAMPLE}(\{1, \dots, K\}, N_C)$   
for  $k$  in  $\{1, \dots, N_C\}$  do  
   $S_k \leftarrow \text{RANDOMSAMPLE}(\mathcal{D}_{V_k}, N_S)$   
   $Q_k \leftarrow \text{RANDOMSAMPLE}(\mathcal{D}_{V_k} \setminus S_k, N_Q)$   
   $\mathbf{c}_k \leftarrow \frac{1}{N_C} \sum_{(\mathbf{x}_i, y_i) \in S_k} f_\phi(\mathbf{x}_i)$ 
```

```
end for
```

```
 $J \leftarrow 0$ 
```

```
for  $k$  in  $\{1, \dots, N_C\}$  do  
  for  $(\mathbf{x}, y)$  in  $Q_k$  do
```

$$J \leftarrow J + \frac{1}{N_C N_Q} \left[d(f_\phi(\mathbf{x}), \mathbf{c}_k) + \log \sum_{k'} \exp(-d(f_\phi(\mathbf{x}), \mathbf{c}_{k'})) \right]$$

```
  end for
```

```
end for
```

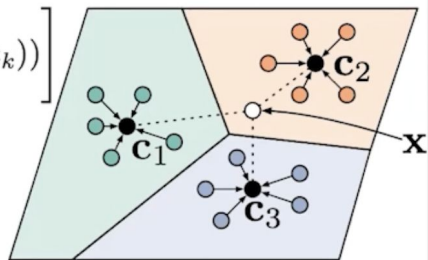
▷ Select class indices for episode

▷ Select support examples

▷ Select query examples

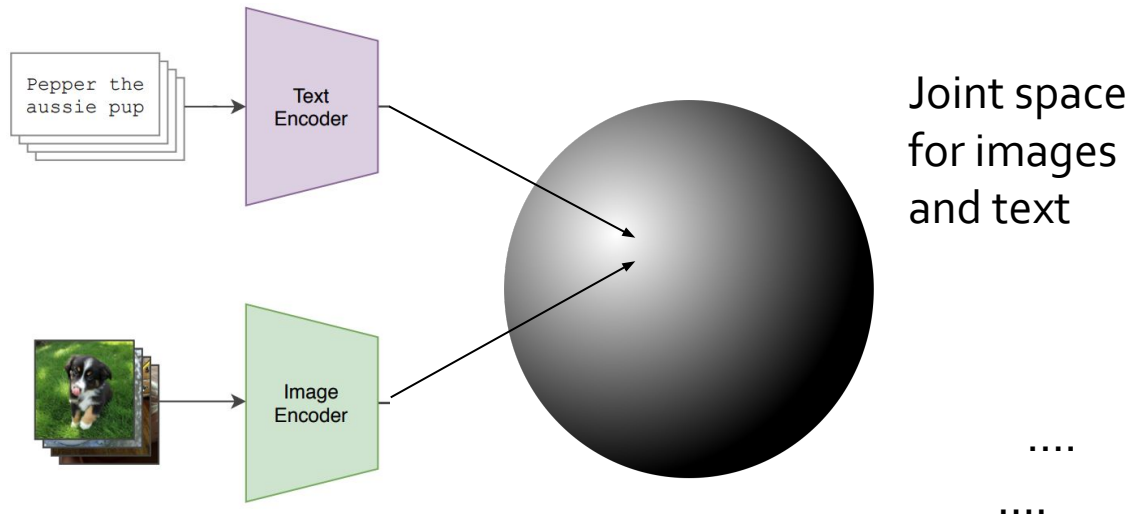
▷ Compute prototype from support examples

▷ Initialize loss



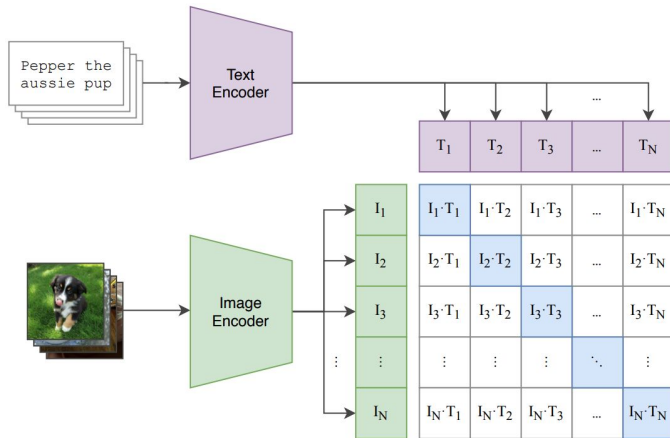
[Snell et al. ICLR17]

CLIP: connecting images and language



[Radford et al. ICML21]

CLIP: connecting images and language

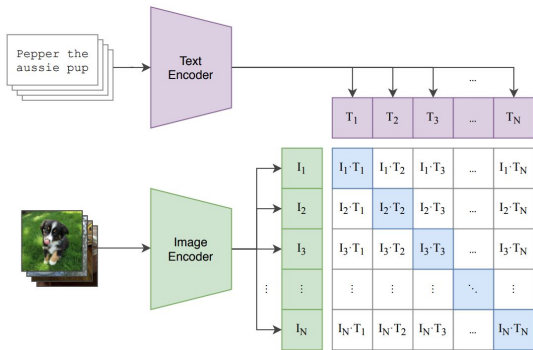


symmetric loss function

```
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

[Radford et al.
ICML21]

CLIP: connecting images and language

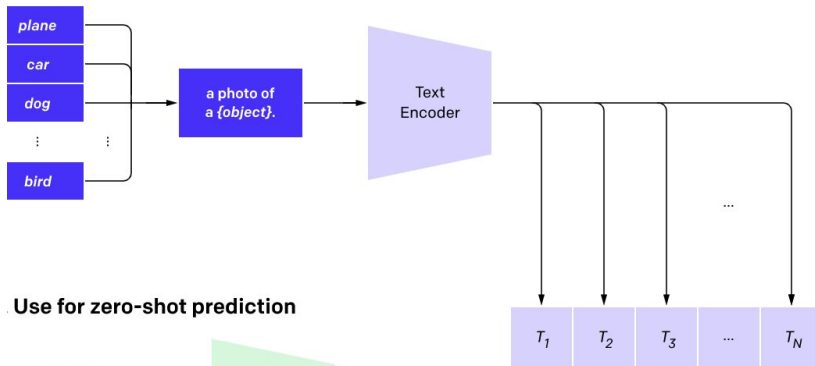


- Text encoder = Text Transformer
- Multiple architectures tried for Image encoder
- No pretraining
- Trained on 400 mln pairs from the Internet for 500,000 text queries

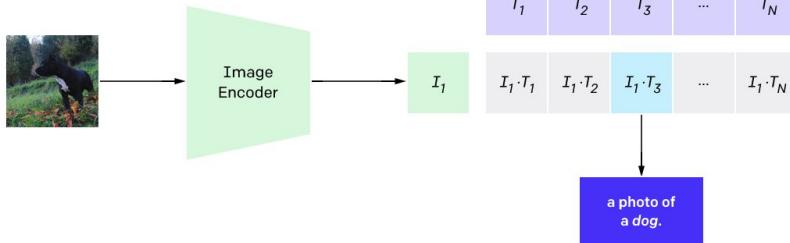
[Radford et al.
ICML21]

CLIP: zero-shot evaluation

Create dataset classifier from label text

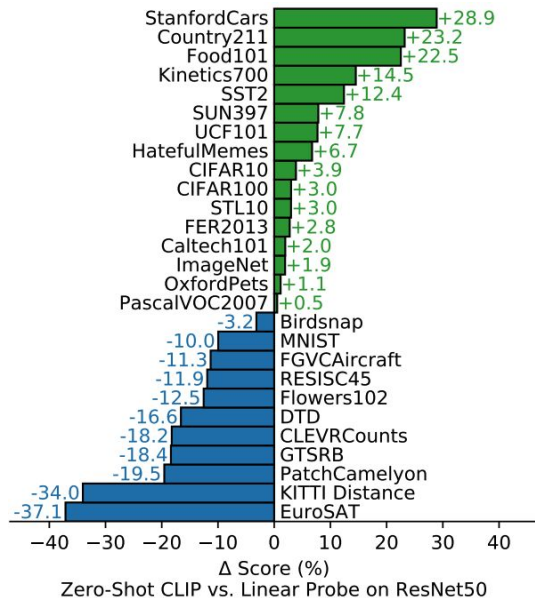


Use for zero-shot prediction



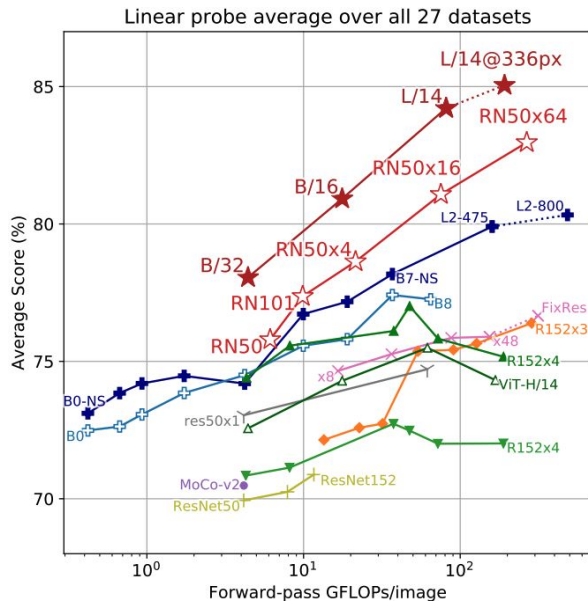
[Radford et al.
ICML21]

CLIP: zero-shot evaluation



[Radford et al.
ICML21]

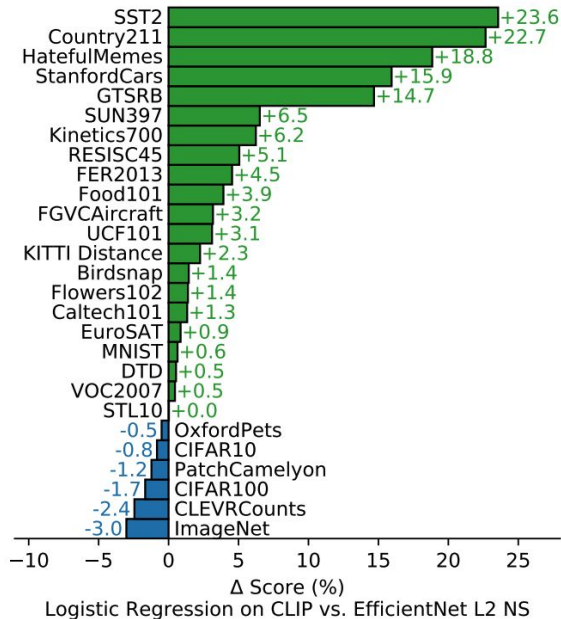
CLIP: linear-probe evaluation



- CLIP-ViT
- CLIP-ResNet
- EfficientNet-NoisyStudent
- EfficientNet
- Instagram-pretrained
- SimCLRv2
- BYOL
- MoCo
- ViT (ImageNet-21k)
- BiT-M
- BiT-S
- ResNet

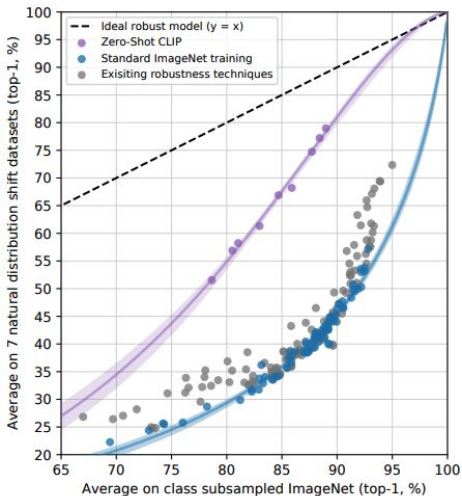
[Radford et al. ICML21]

CLIP: linear-probe evaluation



[Radford et al. ICML21]

CLIP: robustness to distribution shifts



	Dataset Examples						ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet							76.2	76.2	0%
ImageNetV2							64.3	70.1	+5.8%
ImageNet-R							37.7	88.9	+51.2%
ObjectNet							32.6	72.3	+39.7%
ImageNet Sketch							25.2	60.2	+35.0%
ImageNet-A							2.7	77.1	+74.4%

[Radford et al. ICML21]